

Assignment 3 Solutions

NIS3352-01 Modern Cryptography (2)

Problem 1. Let E be the elliptic curve

$$E : y^2 = x^3 + x + 6 \quad \text{over } \mathbb{F}_{11},$$

and let $P = (2, 7)$ and $Q = (5, 2)$. Compute $P + Q$.

Since $x_P \neq x_Q$, the slope is

$$\lambda = \frac{y_Q - y_P}{x_Q - x_P} = \frac{2 - 7}{5 - 2} = \frac{-5}{3} \pmod{11}.$$

Now $3^{-1} \equiv 4 \pmod{11}$, so

$$\lambda \equiv -5 \cdot 4 \equiv -20 \equiv 2 \pmod{11}.$$

Hence

$$x_{P+Q} = \lambda^2 - x_P - x_Q = 2^2 - 2 - 5 = 4 - 7 = -3 \equiv 8 \pmod{11},$$

and

$$y_{P+Q} = \lambda(x_P - x_{P+Q}) - y_P = 2(2 - 8) - 7 = 2(-6) - 7 = -19 \equiv 3 \pmod{11}.$$

Therefore,

$$\boxed{P + Q = (8, 3)}.$$

Problem 2. Let

$$E : y^2 = x^3 + x + 4 \quad \text{over } \mathbb{F}_5.$$

Show that $(3, 2)$ is a 3-torsion point.

Let $P = (3, 2)$. First, check that $P \in E(\mathbb{F}_5)$:

$$2^2 = 4, \quad 3^3 + 3 + 4 = 27 + 3 + 4 = 34 \equiv 4 \pmod{5}.$$

So P lies on the curve.

Now compute $2P$. The tangent slope is

$$\lambda = \frac{3x_1^2 + a}{2y_1} = \frac{3 \cdot 3^2 + 1}{2 \cdot 2} = \frac{27 + 1}{4} = \frac{28}{4} \equiv \frac{3}{4} \pmod{5}.$$

Since $4^{-1} \equiv 4 \pmod{5}$, we get

$$\lambda \equiv 3 \cdot 4 = 12 \equiv 2 \pmod{5}.$$

Thus

$$x_{2P} = \lambda^2 - 2x_1 = 2^2 - 2 \cdot 3 = 4 - 6 = -2 \equiv 3 \pmod{5},$$

and

$$y_{2P} = \lambda(x_1 - x_{2P}) - y_1 = 2(3 - 3) - 2 = -2 \equiv 3 \pmod{5}.$$

Hence

$$2P = (3, 3).$$

But over $\mathbb{F}_5$, the negative of $(3, 2)$ is

$$-(3, 2) = (3, -2) = (3, 3).$$

So

$$2P = -P.$$

Therefore,

$$3P = 2P + P = -P + P = \mathcal{O}.$$

Since $P \neq \mathcal{O}$, the point P has order 3. Hence

$$\boxed{(3, 2) \text{ is a 3-torsion point.}}$$

Problem 3. A pairing $e : G_1 \times G_2 \rightarrow G_T$ is alternating if $G_1 = G_2$ and

$$e(g, g) = 1 \quad \text{for all } g \in G_1.$$

Prove that if e is alternating, then

$$e(g, h) = e(h, g)^{-1}.$$

Since e is a pairing, it is bilinear. Because it is alternating, for any $g, h \in G_1$,

$$1 = e(g + h, g + h).$$

Using bilinearity,

$$e(g + h, g + h) = e(g, g)e(g, h)e(h, g)e(h, h).$$

Since e is alternating, $e(g, g) = e(h, h) = 1$, so

$$1 = e(g, h)e(h, g).$$

Therefore,

$$e(g, h) = e(h, g)^{-1}.$$

Thus,

$$\boxed{e(g, h) = e(h, g)^{-1} \text{ for all } g, h \in G_1.}$$

Problem 4. Assume G_1, G_2, G_T are cyclic. Prove that the following pairing inversion problem is not harder than the discrete logarithm problem in G_T .

Pairing inversion problem: Given $g_T \in G_T$ and $h \in G_2$, find $g \in G_1$ such that

$$e(g, h) = g_T.$$

Let P be a generator of the cyclic group G_1 . Fix $h \in G_2$.

Define the map

$$\varphi_h : G_1 \rightarrow G_T, \quad \varphi_h(X) = e(X, h).$$

By bilinearity, φ_h is a group homomorphism. Hence the image of φ_h is the cyclic subgroup generated by

$$u := e(P, h) \in G_T.$$

If $h = 1_{G_2}$, then $e(g, h) = 1_{G_T}$ for every $g \in G_1$, so the inversion problem is trivial: it has a solution iff $g_T = 1_{G_T}$.

Now assume $h \neq 1_{G_2}$. If g_T lies in the image of φ_h , then there exists an integer x such that

$$g_T = u^x.$$

Finding such an x is exactly a discrete logarithm problem in the subgroup $\langle u \rangle \leq G_T$.

Once x is found, set

$$g = xP \in G_1.$$

Then

$$e(g, h) = e(xP, h) = e(P, h)^x = u^x = g_T.$$

So a solution to pairing inversion can be recovered from one discrete logarithm computation in G_T .

Therefore, pairing inversion is polynomial-time reducible to the discrete logarithm problem in G_T , and hence is *not harder* than DL in G_T .

Problem 5. Let $E[N]$ be the N -torsion subgroup of E , and let (P, Q) be a basis of $E[N]$. Let

$$e : E[N] \times E[N] \rightarrow \langle \xi \rangle_N$$

be an alternating pairing. Show that

$$e(aP + bQ, cP + dQ) = e(P, Q)^{ad-bc} \pmod{N}.$$

Using bilinearity,

$$e(aP + bQ, cP + dQ) = e(P, P)^{ac} e(P, Q)^{ad} e(Q, P)^{bc} e(Q, Q)^{bd}.$$

Since e is alternating,

$$e(P, P) = e(Q, Q) = 1.$$

Also, by Problem 3,

$$e(Q, P) = e(P, Q)^{-1}.$$

Therefore,

$$e(aP + bQ, cP + dQ) = e(P, Q)^{ad} \cdot e(P, Q)^{-bc} = e(P, Q)^{ad-bc}.$$

Equivalently,

$$e(aP + bQ, cP + dQ) = e(P, Q)^{\left| \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right|},$$

since

$$\left| \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right| = ad - bc.$$

Hence

$$\boxed{e(aP + bQ, cP + dQ) = e(P, Q)^{ad-bc}.$$

Problem 6. Suppose that $p > 3$ is an odd prime, and $a, b \in \mathbb{Z}_p$, and that

$$x^3 + ax + b \equiv 0 \pmod{p}$$

has three distinct roots in $\mathbb{Z}_p$. Show that the corresponding elliptic curve group $(E, +)$ is not cyclic.

Let the three distinct roots be $r_1, r_2, r_3 \in \mathbb{Z}_p$. Then

$$r_i^3 + ar_i + b \equiv 0 \pmod{p},$$

so

$$(r_i, 0) \in E(\mathbb{F}_p) \quad (i = 1, 2, 3),$$

because

$$0^2 = r_i^3 + ar_i + b.$$

Set

$$T_i = (r_i, 0) \in E(\mathbb{F}_p).$$

Since $-T_i = (r_i, -0) = (r_i, 0) = T_i$, we have

$$T_i = -T_i,$$

so

$$2T_i = \mathcal{O}.$$

Thus each T_i is a point of order 2.

Hence $E(\mathbb{F}_p)$ contains at least three distinct nontrivial elements of order 2, namely T_1, T_2, T_3 . The set

$$H = \{\mathcal{O}, T_1, T_2, T_3\}$$

is therefore a subgroup of $E(\mathbb{F}_p)$ of size 4, and every non-identity element of H has order 2. Thus

$$H \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

But a cyclic group can contain *at most one* element of order 2. Since $E(\mathbb{F}_p)$ contains three distinct elements of order 2, it cannot be cyclic.

Therefore,

$E(\mathbb{F}_p)$ is not cyclic.

Problem 7. Consider the elliptic curve

$$E : y^2 \equiv x^3 + ax + b \pmod{p},$$

where $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$ and $p > 3$ is prime.

- Prove that if $P = (x_1, y_1) \in E$ has order 3, then

$$3x_1^4 + 6ax_1^2 + 12bx_1 - a^2 \equiv 0 \pmod{p}.$$

- Conclude that there are at most 8 points of order 3 on E .

Assume that $P = (x_1, y_1)$ has order 3. Then

$$3P = \mathcal{O},$$

so

$$2P = -P.$$

Now $-P = (x_1, -y_1)$ has the same x -coordinate as P , hence the x -coordinate of $2P$ must also equal x_1 .

For the doubling formula, let

$$\lambda = \frac{3x_1^2 + a}{2y_1}.$$

Then

$$x(2P) = \lambda^2 - 2x_1.$$

Since $2P = -P$, we have $x(2P) = x_1$, and therefore

$$\lambda^2 - 2x_1 = x_1.$$

So

$$\lambda^2 = 3x_1.$$

Substituting the expression for λ gives

$$\left(\frac{3x_1^2 + a}{2y_1} \right)^2 = 3x_1.$$

Multiplying both sides by $4y_1^2$,

$$(3x_1^2 + a)^2 = 12x_1y_1^2.$$

Since P lies on the curve,

$$y_1^2 = x_1^3 + ax_1 + b.$$

Thus

$$(3x_1^2 + a)^2 = 12x_1(x_1^3 + ax_1 + b).$$

Expanding both sides,

$$9x_1^4 + 6ax_1^2 + a^2 = 12x_1^4 + 12ax_1^2 + 12bx_1.$$

Rearranging,

$$3x_1^4 + 6ax_1^2 + 12bx_1 - a^2 \equiv 0 \pmod{p}.$$

This proves the first claim.

Now consider the polynomial

$$f(x) = 3x^4 + 6ax^2 + 12bx - a^2 \in \mathbb{F}_p[x].$$

It has degree 4, so it has at most 4 roots in $\mathbb{F}_p$.

If $P = (x_1, y_1)$ has order 3, then x_1 must be a root of $f(x)$. For each such x_1 , the curve equation

$$y^2 = x_1^3 + ax_1 + b$$

has at most two solutions for y in $\mathbb{F}_p$, namely y and $-y$.

Hence the number of points of order 3 is at most

$$4 \cdot 2 = 8.$$

Therefore,

there are at most 8 points of order 3 on E .
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